Chapter 7

Reduction Methods of Powder Production

Stanislav S. Naboychenko
Ural Federal University (UrFU), Yekaterinburg, Russia

Chapter Outline

SOLID-PHASE REDUCTION OF METAL OXIDES AND SALTS

In the solid-phase reduction process, metal oxides or metal salts (carbonates, halides, oxalates, and formates) are treated by a reducing agent at temperatures below their melting point. Oxide-reduced powders characteristically exhibit the presence of pores within powder particles and thus are called sponge powders. This sponginess is controlled by the number and size of the pores and accounts for the high green strength as well as the good compactibility and sinterability of such powders. Typically, high reduction temperatures (higher than 60% of the melting point, T_m) facilitate the formation of large intraparticle pores and therefore powders of a specific surface that are required for high compressibility. Parallel with these, elevated temperatures may cause excessive sintering and agglomeration, which cause difficulties with the disintegration of the sintered cake. Low reduction temperatures lead to the production of powders with fine pores, large specific surface areas, and high green strength. But extremely low reduction temperatures ($<0.3T_m$) can readily produce pyrophoric powders.

For metals with very high melting points, such as tungsten and molybdenum, oxide reduction is a benefit partly for economic reasons. Thus, for molybdenum trioxide, because of its high vapor pressure, the reduction is accomplished in two stages to control particle size. The first step, the reduction of molybdenum trioxide to molybdenum dioxide, is carried out at a temperature between 1146 and 1246 K. The second step, the reduction of molybdenum dioxide to molybdenum monoxide, is carried out at around 1446–1646 K.

In contrast to atomized powders, where oxides usually are concentrated on the surface of the particles, oxide-reduced powders, at least when freshly reduced or stabilized against tarnishing, contain most of their residual oxide within the particles.

Theory of the Process [1,2]

The thermodynamics of metal oxide reduction by hydrogen or carbon monoxide are determined based on the analysis of the generalized reaction:

Handbook of Non-Ferrous Metal Powders. https://doi.org/10.1016/B978-0-08-100543-9.00007-5

$$\frac{1}{y}M_xO_y + H_2(CO) \leftrightarrow \frac{x}{y}M + H_2O(CO_2)$$

The latter is analyzed as a set of oxide dissociation reactions

$$M_xO_y \leftrightarrow xM + \frac{y}{2}O_2$$

and gaseous reducer combustion:

$$xB + \frac{y}{2}O_2 \leftrightarrow B_xO_y$$

Below, the corresponding equations for the dependence of the equilibrium constant on the temperature for the reactions of hydrogen combustion ($2H_2 + O_2 \rightarrow 2H_2O$) are given:

$$\lg K_{eq} = \frac{12\,543}{T} - 0.473 \lg T - 3.6 \times 10^{-4}T + 8.1 \times 10^{-6}T^2 - 0.865$$

and carbon monoxide combustion ($CO + 0.5O_2 \leftrightarrow CO_2$):

$$\lg K_{eq} = \frac{14\,250}{T} - 2.65 \lg T - 0.08 \times 10^{-3}T + 100T^{-0.5} - 15.4$$

This process is possible only if the partial pressure of the reducing gas is higher than its equilibrium value typical for the reduction of the oxide of the given metal. An opposite development of the reaction is possible in the case of low concentration of the reducing gas or high concentration of its oxidation products. The reduction system is a three-component and three-phased one (oxide, metal, gas), therefore there are two degrees of freedom. The equilibrium constant (K) of the reduction reaction is determined by the ratio of partial pressures of the reducing gas and the product of its oxidation. The equilibrium gas composition is determined by its pressure and temperature. Only temperature influences the equilibrium constant (K) at low pressures of the gas; therefore, the reaction proceeds without changing the number of gas moles. If the reaction constant is $K \leq 1.0$, the oxides are difficult to reduce and vice versa. The stronger the metal bond with oxygen, the higher the oxide formation heat and the more positive the thermal effect of its reduction reaction.

The utilization factor of the gaseous reducer (η%) is determined by the equilibrium constant of the reaction

$$\eta = \frac{K}{1+K} \cdot 100$$

and can never achieve 100%. The lower the value of the equilibrium constant, the higher the hydrogen consumption; the utilization factor of hydrogen increases with its dehumidification.

The partial pressure of the metal in the gas phase enters into the equation for the reaction equilibrium constant by the reduction of metal compounds with high vapor pressure. The decrease of the total pressure and the increase of temperature and reducing gas concentration facilitate the development of similar processes. If the equilibrium metal pressure is higher than its saturated vapor pressures, the production of metal by condensation is possible. If the reduction reaction is endothermic, the opposite reaction (i.e., oxidation by carbon dioxide or water vapor) is possible during the cooling of the gaseous mixture together with the condensation of metal vapors. The formation of oxides on metal particle surfaces is possible in the presence of oxidizers in the gaseous phase. This complicates the coalescence of metal particles.

Kinetics and Mechanism of the Process

Reduction processes are developed in accordance with the Baykov principle on the consecutive transformations of all chemical compounds formed in the given system. The reactions of indirect reduction are special cases of topochemical reactions; they proceed on the interphase boundary [3].

The kinetics of solid-phase reduction depend on the process parameters (temperature, pressure, reducer type, and mass transfer conditions) as well as on the physical and chemical properties of the initial raw material (structure defects, dispersivity, and specific surface). According to the adsorption-autocatalytic theory, the main stages of the process are:

1. Supply of a gaseous reducer to the surface of the processed material.
2. Adsorption of the reducing gas molecules on active centers of the solid surface.
3. Recombination of the reducing gas molecule under the influence of crystal lattice forces that is accomplished by the formation of an activated form at the atomic level.

4. Diffusion of reducing gas atoms through pores and cracks along the defects of the crystals lattice, and filling of electron vacancies.
5. Chemical interaction (electron interchange, nucleation, and growth of metallic phase crystals); the latter accelerates the adsorption and dissociation of molecules and provides the autocatalytic nature of further reduction due to high surface activity.
6. Desorption of gaseous oxidation products of the reducer through the layer of the metallic phase.
7. Diffusion of ions of metal and oxygen; electron transfer inside the crystalline phase.

The supply of the reducing gas (an act of external diffusion) in conditions of its excess consumption, purity, and optimal feed rate does not retard the process. The process rate limited by the adsorption stage depends on the pressure of the reducing gas (p_g) and is described by one of the following equations:

$$\nu = k\frac{bp_g}{1+bp_g}$$

$$\nu = k \cdot p_g^n$$

where k is the rate constant; and b and n are experimental coefficients, which usually are not higher than 1.0.

For reversible reactions:

$$\nu = k\frac{p_g - p_g^1}{p_{pr}^n}$$

where p_g and p_{pr} are partial pressures of the reducing gas and reaction product; and p_g^1 is the equilibrium pressure of the reducing gas.

Reducing gas comes to the reaction zone in accordance with laws of internal diffusion and its supply is increased with the growth of the gas pressure and the lowering of the temperature. Adsorption of the reducing gas molecule is accompanied by the transmission of electrons in the oxide lattice. Adsorption mechanisms are different for oxides of hole (p-type) semiconductors with cation vacancies (Cu_2O, NiO) and electronic (n-type) semiconductors (ZnO, CdO and Ti_2O) with cations in interstitial spaces or with vacancies in the oxygen lattice. Owing to its insignificant affinity to electrons, hydrogen is a donor in the case of adsorption of its molecule on the oxide surface; in this case, a positive charge containing H^+ ions arises.

As the process develops, a metallic phase layer is formed that complicates diffusion conditions and decelerates the reduction rate. The negative influence of the metallic film is weaker when hydrogen is used because its molecule size is smaller than that of carbon monoxide while the diffusion coefficient is larger ($D_{H_2} = 3.74\ D_{CO}$). The elevated activity of hydrogen especially appears in the region of lower temperature. Diffusion of metal ions and electrons to the boundary layer accomplishes the metallic phase nuclei:

$$n\mathrm{M}^{2+} + 2n\mathrm{e}^- \leftrightarrow n\mathrm{M} \rightarrow (\mathrm{M})_n$$

A chemical reaction includes three sequential stages: chemisorption of a reducing gas molecule on the reaction surface; separation of an oxygen ion from the oxide lattice and its interaction with activated atoms (molecules) of the reducer that is completed by metallic phase formation; and adsorption of gaseous products of the reduction.

In the beginning stage, the reduction develops slowly due to different rates of reducing gas chemisorption at centers of the oxide surface, which differ by their activity. In the beginning stage for p-type semiconductors, all holes are involved and adsorbed H_{abi}^+ ions are formed:

$$H_{2(g)} \leftrightarrow 2H_{adi}^+\left(M_\beta^{2-}\right)$$

The free electrons are formed in the boundary layer:

$$0.5\,H_2(g) \leftrightarrow H_{adi}^+ + e$$

Reduction develops with the filling of cation vacancies M_β^{2-}

$$2H_{adi}^+ + MO + M_\beta^{2-} \leftrightarrow H_2O_g + M$$

and further with electron usage:

$$2H^+_{adi} + MO + 2e \leftrightarrow H_2O_g + M$$

For *n*-type semiconductors, the reduction process includes the following stages:

- Chemisorption of hydrogen and ionization of atoms:

$$H_{2(g)} \leftrightarrow H_{2adi} \leftrightarrow 2H_{adi} \leftrightarrow 2H^+_{adi} + 2e$$

- Connection of free electrons by metal oxide ions.
- Elimination of oxygen with formation of vacancies on the surface.

$$H_{2(g)} \leftrightarrow H_2O_g + O^{2+}_{\beta} + 2e$$

and collection of holes on a metal oxide surface.

- Metal ion formation in interstitial spaces of (x) lattice.

$$H_{2(g)} + MO \leftrightarrow H_2O + M^{2+}(x) + 2e$$

In the case of a multistage mechanism of the reduction process, which includes a series parallel formation of the lower oxides, the reaction rate is determined by the lowest rate among these stages. A zonality process of an autocatalytic type is observed, specified by the different catalytic activity of the formed phases.

Transformation of the lattice of the higher metal oxide into the lattice of the lower oxide or metal lattice occurs in accordance with the principle of phase structure succession. With decreasing temperature, the induction period becomes longer and the speed of interphase boundary movement decreases. Metal ions as well as oxygen atoms can take part in diffusion stages; the limiting stage is determined by the ratio of their diffusion rates. The diffusion is carried out through crystal lattice volume and along the grain boundaries.

In real conditions, due to the use of elevated temperatures, the rate of crystal chemical transformation is much higher than the rate of reducer supply or the elimination of gaseous reaction products, that is, the reduction process flows in the diffusion area.

Rostovtsev [4] suggested a kinetic equation, which described the diffusion-kinetics range of the reduction process:

$$\tau = \frac{q}{\Delta N}\left[\begin{array}{c}\dfrac{1}{k}\ln\dfrac{e^{S}(S-1)+e^{-S}(S+1)}{e^{S\cdot\xi}(S\cdot\xi-1)+e^{-S\cdot\xi}(S\cdot\xi+1)} \\ +\dfrac{R^2}{D_{2e}}\left(\dfrac{1-\xi^2}{2}-\dfrac{1-\xi^3}{3}\right)+\dfrac{4R^2\left(1-\xi^3\right)}{3D_{1e}\left(\sqrt{Re}+4\right)}\end{array}\right]$$

where τ is the reduction duration, s; q is the gas mass necessary for full reduction of 1 cm^3 of raw material, mole/cm^3; R is the radius of a spherical piece, cm; ΔN is the reducer excess over its equilibrium concentration; k is the constant of the crystal transformation rate; $S = R\sqrt{k/D_{1e}}$ is the dimensionless simulation criterion; D_{1e} and D_{2e} are the coefficients of internal diffusion in volumes of an initial material particle and a particle after reduction, mol/(cm s); ξ is the dimensionless parameter; $\xi = (1-\omega)^{1/3}$, where ω is the reduction degree, in unity fraction; and Re is the Reynolds's criterion.

As the criterion S decreases, the process shifts to the kinetics area; in the transition area S ranges from 10 to 50; diffusion limits become determinative at high values of this criterion.

The following guidelines are useful for metal powder production practice:

- Some impurities contained in the gaseous reducer can be adsorbed on active surface centers more rapidly and capably, thereby reducing the front of the productive reaction. Therefore, it is necessary to use pure gases with prior drying.
- Increasing particle surface and dispersivity, decreasing the thickness of the material layer, and enhancing mass transfer conditions (increasing the active concentration and speed of the reducing gas, and layer stirring) are required for process acceleration by its behavior in the diffusion area.
- Temperature monitoring, excluding subsmelting of particles of newly reduced metal and retaining the gas-permeability of the treated layer.
- The oxide reduction rate increases in the presence of a number of impurities in the basic material. The degree of oxide structure disorder increases by substitution of the matrix phase cation particles, for example, by selenium and tellurium cations. This leads to a number of vacancies and holes in p-type oxide-semiconductors increase that simplifies ion diffusion, accelerates chemisorption of the reducing gas, makes easier crystal transformations.

- Oxides of alkali metals interact with microimpurities (sulfur, phosphorus, and arsenic) with their transfer into water-soluble compounds, which are eliminated in the process of cake washing that increases the powder purity.
- The size of powder particles significantly depends on the initial product dispersivity.
- Powder particle size appreciably depends on the reduction temperature; the finer powders are produced by the reduction of chemically precipitated salts at a moderate temperature (573–773 K), heightened reducer consumption, and limited duration.
- To increase the gas permeability layer as well as exclude powder sintering and grain coarsening, it is useful to granulate the charge by the injection of 1%–3% binders (alcohols, organic acids, or chlorides of alkali-earth metals), which then are sublimated at reduction temperatures.
- Powder purity greatly depends on the composition of the basic raw material, as only impurities forming volatile compounds (C, As, S, O, and Cl^-) can be eliminated in the reduction process.

Feedstock and Reagent Reducers

Reducers are gaseous (hydrogen, carbon monoxide, converted natural gas, and dissociated ammonia), solid (graphite, charcoal, and other carbon-bearing reagents with limited content of sulfur, phosphorus, and arsenic), or combined reducers (for example, carbon monoxide and charcoal). The gases are produced in dedicated plants and oxygen and water vapor are removed.

Hydrogen is the most effective reducer. Four industrial methods of hydrogen production (conversion of natural gas and electrolysis of aqueous solutions) are well known and readily available commercially: A, by electrolysis of aqueous solutions; B, by the iron-steam method and by the interaction of ferroslag with the alkaline solution; C, by chloride electrolysis; and D, by steam conversion of hydrocarbon gases (Table 7.1).

Hydrogen is purified from oxygen in vertical furnaces filled with a catalyst (copper sponge, chromium-nickel powder, or an iron-nickel-copper mixture at 673–723 K. The dehydration is carried out in columns filled by silica gel and cement; the residual moisture content is not higher than 20 mg/m^3, which corresponds to a dew point about 228 K. Silica gel is periodically regenerated at 450 K. Deeper dehydration of hydrogen, giving a residual content of 2–4 mg/m^3 and removal of carbon dioxide, is achieved by passing the gas through columns with granulated potassium hydroxide. To exclude the formation of explosive mixtures, gas is supplied with a surplus to the stoichiometry (the efficiency is below 15%–20%), therefore the discharge gases are burned or directed to regeneration. The latter provides cleaning from dust, moisture, and oxygen. Methods of diffusion separation on palladium-containing membranes and of short-cycled adsorption under a pressure with blowing off adsorbent regeneration are used in case of need to free hydrogen from CH_4, CO, H_2S and N_2 and producing particularly pure hydrogen. The former is implemented for devices with low capacity while the latter is for devices with hydrogen consumption amounts of 1000 m^3/h and higher.

TABLE 7.1 Physical and Chemical Properties of Technical Hydrogen

	Brand					
			C		D	
Properties	**A**	**B**	**1**	**2**	**1**	**2**
Chemical composition						
Hydrogen content, vol%, min	99.8	98	98.5	97.5	97.5	95
Impurties: vol%, max						
Oxygen	0.2	0.3	0.3	0.5	0.2	0.2
Carbon monoxide	–	0.3	0.2	0.3	0.5	1.2
Carbon dioxide	–	0.5	9.2	0.3	0.5	1.0
Water steam content (298 K, 0.1 MPa) (g/m^3)	25	25	25	25	25	25
Max: transportation via gas conduits and storage in rubber-textile gasholders						

The kind of adsorption under pressure technology, called "pressure swing adsorption (PSA)," has become state-of-the-art technology for the production of high purity H_2 (99.995+%) from a feed gas containing 60%–90% H_2. It is used for >85% of global hydrogen production [5–8].

The basic principle of an H_2 PSA process for these applications is relatively simple. The bulk and dilute impurities present in the feed gas are adsorbed by passing through a column packed with one or more adsorbents in order to produce the pure H_2 product gas at feed pressure. The impurities are then desorbed by lowering their superincumbent partial pressures inside the column in order to produce an impurity-rich gas. Though simple in principle, a practical PSA process can be complex, consisting of the adsorption and the desorption steps in conjunction with a variety of complementary steps that are designed to improve the H_2 product's purity and recovery, and to reduce the adsorbent units. Thus, a PSA process involves a series of sequential nonisothermal, nonisobaric, and unsteady-state steps operated in a cyclic steady-state fashion using multiple parallel adsorption columns.

The two most commonly used gas sources for H_2 production are steam-methane-reformer off gas (SMROG) after it has been further treated in a water-gas-shift (WGS) reactor, and refinery off gases (ROG) from various sources. The most frequently used H_2 PSA processes are used for the production of high-purity H_2 from the natural gas. A popular design called the polybed process contains nine sequential steps, described by Sircar [5]. The adsorbers are packed with a layer of activated carbon in the feed end and a layer of zeolite in the product end. The process was originally used to produce high-purity H_2 at high H_2 recovery from SMROG.

Fig. 7.1 is a schematic box diagram for the most commonly used steam methane reforming (SMR)–water-gas shift (WGS)–PSA route of H_2 production [6,9]. Steam and natural gas in the ratio of 2.5–6.0 react ($CH_4+H_2O \leftrightarrow CO+3H_2$) at a high temperature (1023–1173 K) and pressure (0.4–3.0 MPa) over a nickel catalyst. In an endothermic SMR reactor, the effluent gas comprising (in vol%) (70–72)H_2, (6%–8%)CH_4, (8–10)CO, and (10–14)CO_2 (dry basis) is cooled to 573–673 K (steam-produced) and subjected to the exothermic WGS reaction ($CO+H_2O \leftrightarrow CO_2+H_2$), cooled again, and purified in a multicolumn PSA unit. The H_2 product contains 99.999 H_2 (<15 ppm CO_x) and the H_2 recovery volume from the PSA unit is typically 70%–95%. The PSA waste gas is recycled as fuel to the reformer, with fresh natural gas providing the balance of the fuel.

Elseviers et al. [7] reported that the current generation of polybed PSA systems is characterized by advancement induced by the availability of improved computational capabilities of personal computers and advanced adsorption simulation programs. A new generation of ultrahigh performance adsorbents allows further reduction in the installed adsorbent volume per ton of hydrogen recovered without sacrificing purity or recovery.

According to Linde's group comp. Data [8], the capacities range from a few hundred Nm^3/h to large-scale plants with >400,000 Nm^3/h that are commercially available. The hydrogen product meets every purity requirement up to 99.9999 mol

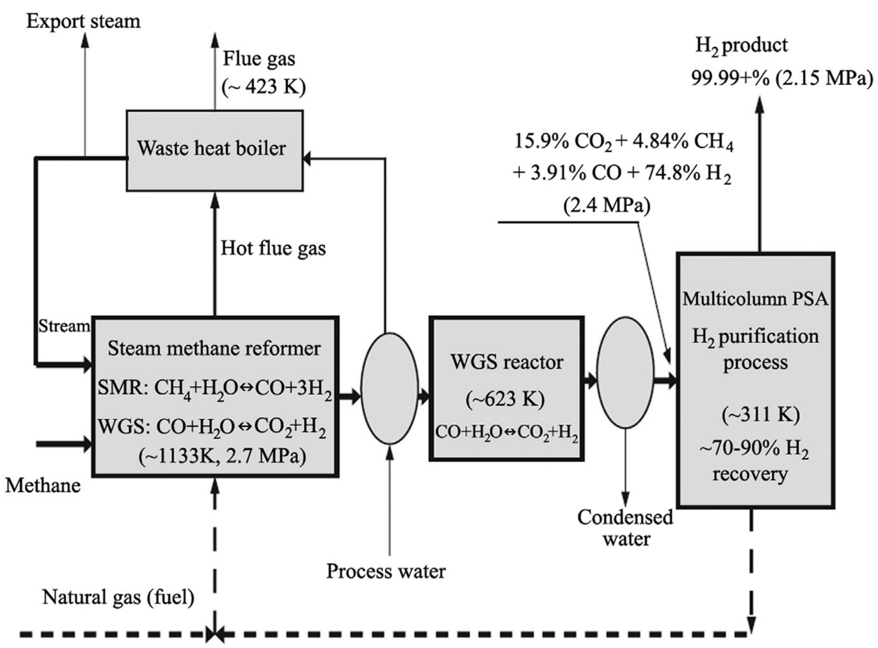

FIG. 7.1 Conventional steam methane reforming (SMR) route for hydrogen production.

%. Final desorption and regeneration is performed at the lowest pressure of the PSA sequence. Highly pure hydrogen obtained from an adsorber in the provided purge step is used to purge the desorbed impurities into the tail gas system. Before restarting adsorption, the regenerated adsorber is pressurized again. This is accomplished in the pressure equalization step by using pure hydrogen from adsorbers presently under depressurization.

Hydrogen is an explosive gas that requires special permissions according to safety measures. When choosing structural steels to manufacture devices that operate in hydrogen atmospheres, it is necessary to take into account that the formation of volatile hydrides with impurities contained in the steel (sulfur, phosphorus, and carbon) is possible at high temperatures. Recrystallization of the metal structure and a decrease of strength properties are possible during long-term operation.

Endothermic gas is produced by incomplete combustion ($\alpha = 0.25$–0.3) of the natural gas at 1300 K. It has the following composition: 38–40 vol% H_2, 18–20 vol% CO, 1–2 vol% CO_2, balance N_2; the dew point is in the range of 268–273 K. Exothermic gas is produced by means of combustion of the natural gas with somewhat higher air consumption.

Dissociated ammonia is produced from liquid ammonia by means of its vaporization, heating and dissociation, cooling, and drying. The gas has the following composition: 75 vol% H_2 and 25 vol% N_2; dew point is in the range of 213–230 K without afterburning, and with afterburning, 4–20 vol% H_2 and nitrogen balance.

Converted natural gas is produced by processing of natural gas by steam at 1170–1370 K over a catalyst. A volume of 4 m^3 of reaction gas is produced from 1 m^3 of natural gas. Conversion furnaces with a capacity of 1000 m^3/h are used for the process. The gas has the following composition: 75–76 vol% H_2, 22–23 vol% CO, 1–2 vol% CO_2, 1–1.5 vol% H_2O, 1.5 vol % N_2, max, and 0.4 vol% CH_4; the dew point is 293 K.

The furnace is purged by nitrogen before the introduction of the explosive and combustible gases into the furnace and in the case of the formation of a neutral atmosphere or as a reagent for nitration of refractory metals and alloys. Nitrogen is produced by cryogenic separation of air; its properties are given in Table 7.2. Nitrogen costs about three times less than hydrogen. Argon is used as a neutral atmosphere in the production of extreme purity metals.

In industrial practice, coke, anthracite culm, charcoal, and carbon black are used among the solid reducers. Coke is produced by means of heat treatment of special coals or petroleum. It is an effective reducer, comprising 93–98 vol% carbon and up to 5 vol% hydrogen in the form of hydrocarbons. However, depending on the composition of the initial raw material, coke includes up to 2%–4% sulfur and 0.5%–1.0% ash, that is, impurities, which lower the quality of the metal powder produced. Anthracite culm is produced by means of anthracite heat treatment. It contains up to 75% active carbon, but also a considerable amount of ash (10%–20%) and sulfur (3%–3.5%). Charcoal is the solid reducing agent with the lowest sulfur content, but it has a high ash content (2%–2.5%) and is also high in volatile matter (17%–20%).

Carbon black is an ultrafine product of hydrocarbon heat treatment: its chemical activity is determined by its large specific surface (up to 15–18 m^2/g). Lampblack is produced by the combustion of liquid petroleum products in wick burners with air deficiency at 1470 K. It contains 0.1% ash and 0.5% moisture maximum. Gas black is produced by the combustion of natural gas with an air deficit. It contains 0.06% ash and 2.5% moisture, maximum.

TABLE 7.2 Commercially Available Nitrogen

		Gaseous and Liquid Nitrogen			
		Grades			
Properties	**Technical Gaseous Nitrogen**	**Extra Pure**	**I grade**	**II grade**	**III grade**
Chemical composition					
Nitrogen content (vol%, min)	99.994	99.996	99.6	99	97
Impurities content (vol% max)					
Oxygen	0.005	0.001	0.4	1.0	3.0
Water (293 K, 0.1 MPa)	0.005	0.005	0.07	…	…
Hydrogen	…	0.001	…	…	…
CO, CO_2, C_nH_m, total	–	0.001	…	…	…
Dew point (0.1 MPa), K	210	210	230	…	…

When coal-based solid reducers are used, it is necessary take the following into account:

- Process parameters depend significantly on purity, porosity, dispersity, and surplus concentration of the reducer.
- In the case of an excess of the reducer and high temperature (especially up to 1470 K), there is a high probability of carbon monoxide formation by oxidation of carbon by the carbon dioxide generated.
- The kinetics of carbon monoxide formation determine the rate of the total reduction process.
- The carbon monoxide yield increases by lower pressure (the reaction proceeds with the increase of molar volumes).
- Decomposition of carbon monoxide with solid-phase formation (soot carbon) is possible in a catalyst's presence (for example, just formed powders of the iron subgroup).

Gaseous reducers are preferable to solid-phase ones because they are more active and contaminate the powder to a lesser degree. Hydrogen has the greatest commercial importance for metal powder production.

POWDER PRODUCTION

The flow sheet of powder production by solid-phase reduction of metal compounds includes three main stages: preparation of the raw material (blending, milling, drying, and degreasing), reduction, and powder processing (milling, classification, and mixing). Preparation of the raw material may also include an oxidation process of the metallic part, as in the copper-oxide process for the manufacture of a typical starting material for both the red cuprous oxide, Cu_2O, and the black cupric oxide, CuO, for producing pure copper powder. The oxidation of air atomized, water-atomized, or shotted copper leads to a radical change in powder particle shape and thus enhances the control of various engineering properties of powders. More detailed information is found in Section D, which is devoted to the methods of individual nonferrous metal powder production.

For producing PM alloys and composites, the initial components are thoroughly mixed in the required proportion and finely milled if necessary. More homogeneous mixtures are produced by chemical precipitation. Solid-phase reduction is carried out in electrical muffle furnaces of discontinuous or continuous operation. Furnaces of continuous operation (drum, push type, conveyor, and walking beam) are the most popular. These are productive, mechanized devices with moderate energy consumption, but they are more complicated in design and in operation compared to furnaces of discontinuous operation.

Push Type Furnaces

The furnaces (Fig. 7.2) have metallic, water-cooled housing lined internally with layers of fire brick. The metal-oxide powder is filled into flat metal boats, which are loaded and unloaded automatically. This results in a static powder layer throughout the whole reduction process. Several heating zones along the furnace length are maintained. Typical reduction temperatures range from 873 to 1373 K.

The water vapor formed during reduction is carried away by the reducer gas flow. The flow of reaction gas is usually in the countercurrent direction. By variation of reduction parameters such as temperature and powder layer thickness, the particle size can be controlled over a wide interval. For instance, in the case of tungsten powder, the particle sizes can be controlled in the range between 0.1 and 100 μm. The operation of the furnace is complicated by fast wear of the guides used for boat movement and by the danger of their jamming.

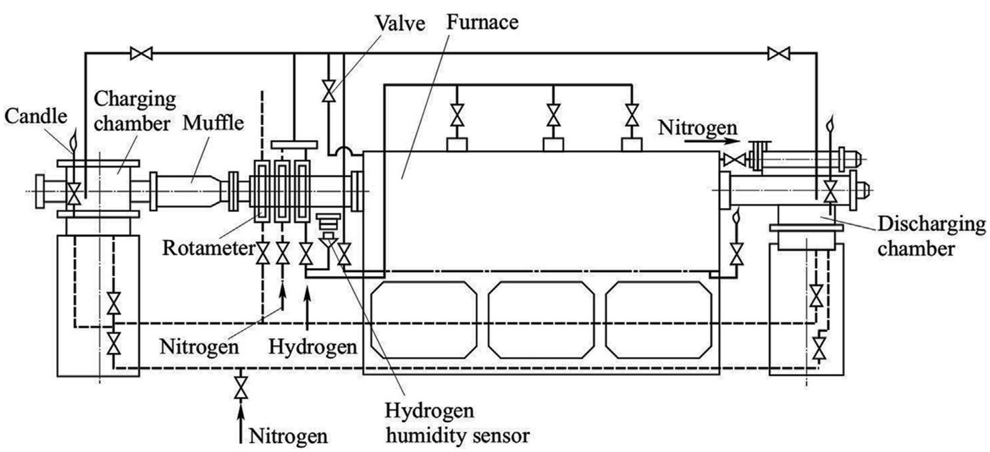

FIG. 7.2 Tying scheme for push-type furnace working on hydrogen.

Drum Furnaces

This type of furnace, called also a rotary furnace, is convenient for use in automatic lines. The metal oxide is directly introduced into the rotating furnace tube. As a consequence, there is no static but a dynamic powder layer. The depth of the layer is influenced by the feed rate, rotational speed, incline, and lifters inside the tube. The temperature range is comparable to that of push-type furnaces and the hydrogen flow direction is also, in most cases, countercurrent. As regards changes of particle size of the metal powder produced, the rotary furnace is not as flexible as the push-type furnace. They have high heat efficiency and easy pressurization while providing heat uniformity and good contact of the material with the reaction gas. But there is a danger of material nodulizing and adhering to the internal wall of the muffle (in cases of wet material processing and local submelting). In addition to these, there is appreciable erosive wear, additional energy consumption (for the rotator drive), more complicated pressurization of powder charging and discharging units, and increased dust formation.

Conveyor Furnaces

These continuous furnaces are used for chemical-thermal treatment and sintering of powders at temperatures not higher than 1423 K. The furnace operation controls some properties, particularly particle size and shape, apparent density, and green strength of the powder production. For example, roasting of air-atomized, water-atomized, or shotted copper is accomplished in belt conveyor furnaces. In commercial practice, oxidation or roasting copper powder is normally done in air at temperatures above 923 K. Reduction of the copper oxide thus produced is generally accomplished in a continuous belt furnace.

In a typical operation, the powder is transferred into the charge bin of a stainless steel mesh belt electric furnace. To prevent the powder from falling through the belt, a continuous sheet of high wet-strength paper covers the belt and then the powder is loaded onto the paper. A roller compresses the powder to improve heat transfer. As it enters the furnace, the paper burns, but not before the powder has sintered sufficiently to prevent it from falling through the belt. The gas enters the furnace from the discharge end and, because it is refrigerated, aids the cooling of the powder cake. By varying the furnace temperature between 753 and 1033 K and the time of exposure, considerable variation can be made in the content of fines, apparent density, and shape. To exclude powder oxidation, it is cooled down to 300–330 K in a reducing gas atmosphere and on completion of the furnace operation, the cake is broken and is ready for grinding. The reliability of the furnace operation is determined by the lifetime of the conveyor belt, which is operated in conditions of elevated temperatures, mechanical loads, and a corrosive environment.

Fluidized Bed Furnaces

The reduction process is accelerated 3–4 times when using fluidized bed furnaces due to increasing the contact area between the powder and the gas-reducer. However, the reduction operations are more difficult to control than reduction in a conveyor furnace. Along with this, dust entrainment and reducing gas consumption enlarge and, as a consequence, the cost of purification and regeneration of the forming gas phase increases.

When using calcium carbide or hydride as the solid reducer, the initial material is thoroughly dried and ball milled together with the reducer to <100 μm particle size. It is then thoroughly blended and processed in the furnace at a required temperature that provides the reduction of the metal or metal compound oxides. A dispersing agent (calcium oxide or sodium chloride) is injected to decrease the agglomeration of metallic particles. Discharge of the resulting cake is fulfilled by a water jet (with simultaneous cooling and removal of excess calcium compounds). The pulp is processed in hydrocyclones for lime separation, acidulated by hydrochloric acid, and then washed free from the calcium chloride formed. The powder is then dewatered in a centrifuge and dried in a nonoxidizing atmosphere.

The extra pure metal powders are produced by reacting low-boiling metal halides (MH) (chlorides and fluorides of molybdenum, rhenium, tungsten, tantalum, and niobium) by hydrogen (Fig. 7.3). A mixture of hydrogen and inert gas, usually argon (in ratio H_2 to Ar–1.5 to 2.0), is supplied in the amount that provides a stable fluidized bed in the furnace. Hydrogen consumption exceeds the theoretically required H_2 to MH ratio several times; therefore, waste gases are regenerated in order to utilize both the hydrogen and hydrogen chloride (hydrogen fluoride) that are returned to the technological process. The total degree of reduction achieves 98–99%; the precipitation takes place on seed particles 10–50 μm in size, which are produced by fine powder milling. Particle sizes of the powder produced range from 200 to 500 μm; the particles are spherical and the grain texture is dense.

Reduction processes are significantly accelerated and the produced powder is distinguished by ultrafine dispersity (30–70 nm) by using low-temperature plasma. Hydrogen or convertible natural gas is used as a reducer.

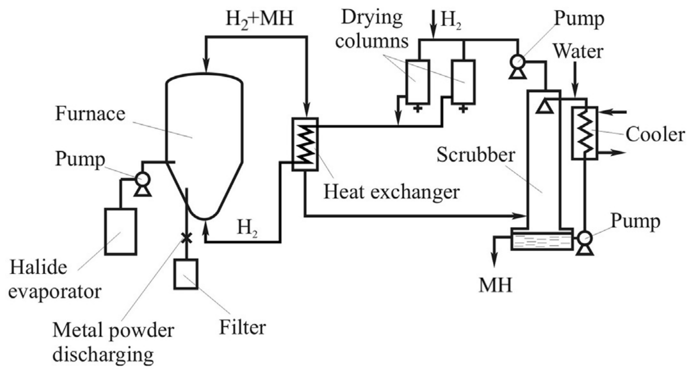

FIG. 7.3 Scheme of device circuit for halide metal (MH) reduction by hydrogen in a fluidized bed furnace.

SAFETY ENGINEERING

The main hazards in metal powder production by reduction processes are dust formation, thermal and electrical action, and explosion risk.

The building is constructed in conformity with the standards of technical design (discussed in Chapter 27). The standards cover adequate ventilation, explosion-proof lighting, lightning protection, and additional safety exits [10].

Safety-critical conditions for operation are the following:

- Good working order of the equipment (explosion-proof and fire-proof design, gas tightness, earthing, protective cover of rotating units, and performance of local ventilation according to standard requirements).
- Safe methods of process conducted by experienced personnel (mechanization, remote tests and control, correct application of personnel protection and firefighting means, dust handling, hygiene and sanitary precautions, and so on).
- Maintenance of design specifications concerning thermal and water supply, the arrangement of freight flow and the usage of leak-tight packages for powders, acoustic and light alarms, and warning labels.

The explosive data of burning gases mixed with air are shown in Table 7.3 [11,12]. Design, installation, and gas tubing operation have to meet regulations for installation and safe operation for combustible, toxic, and condensed gases [11]. The European Development Centre, in Sheffield, the United Kingdom, has Material Safety Data Sheets (MSDS) for industrial gases and gas mixtures available in published form [13].

Reaction gas is supplied either directly to the cold furnace or after purging with inert gas. Gas pressure ranges from 0.1–0.2 MPa; an acoustic alarm is activated if it fails.

TABLE 7.3 Combustion Data of Burning Gases Mixed With Air

	Combustion Limits, vol%		
Gas	**Lower**	**Upper**	**T_B (K)**
Ammonia	14.0	33.0	1053
Butane	1.5	8.5	748–823
Carbon oxide	12.5	74.2	913–926
Hydrogen	4.0	74.2	853–863
Methane	5.0	15.0	923–978
Propane	2.4	9.5	791
Exogas*	17.5	87.5	…
Endogas*	8.5	80	…

Ignition temperature (T_B) depends on gas composition.

The furnaces are equipped with special chambers with double doors from the charging and discharging sides to prevent the ingress of atmospheric air. Automatic temperature control is provided. The temperature shall not exceed 318 K on the external surface of the thermal insulation of furnace screens.

Before the arrangement of fuel spray, the absence of flap while inflame (test-tube trial) and flame combustion stability are controlled. In the case of a sudden failure of the gas supply and fuel spray disappearance, it is immediately necessary to close the output gas fitting and stop the furnace charging and discharging processes. The line of gas supply to the furnace is blocked quickly in the case of gas inflammation.

The speed of gas movement through the suction tubes is chosen to prevent dust precipitation in them. The gases are freed from dust in devices equipped with explosive valves. During furnace repair, access inside is permitted only after cutting off all services, cooling below 313 K with a constant rate, blowing by steam or inert gas that contains 4 vol% O_2 maximum, and water washing of precipitated powder if necessary.

METALLOTHERMY

This process is the reduction of oxides and salts of metal by another metal that has a greater affinity for the metalloid included in the composition of the initial compound. Calcium, magnesium, aluminum, and sodium are mostly used for the reduction. The reducer is chosen according to its activity, availability, volatility, inability to form compounds with the produced metallic powder, and the ease with which it can be removed from the final product. Calcium hydrides or carbides are kinds of reducers. Metallothermy excludes metal powder contamination by carbon. It is very important for the production of precise materials and units made of them. However, the more complicated the reduction reaction, the more probable the contamination of the powder by the metal reducer.

The metallothermy reaction equilibrium is determined by the equality, for example, of oxygen pressures by dissociation of the initial metal oxide and the produced oxide of the metal reducer. The metallothermy process proceeds in autogenous mode if the heat of the reaction is not <2300 kJ per 1 kg of the reaction mixture. In metallothermic reduction, a cake is usually produced; its finishing to commercial powder requires the usage of milling, screening, and uniform mixing.

CHEMICAL REDUCTION

Chemical Precipitation from Solutions

Chemical precipitation of metal as a powder is an oxidation-reduction reaction, which can proceed in an aqueous as well nonaqueous medium [14]. Chemical precipitation from an aqueous medium permits production of all metals capable of coexistence with water: cadmium, thallium, cobalt, nickel, tin, lead, bismuth, rhenium, indium, copper, and noble metals. However, thermodynamically possible reactions are often impracticable due to their extremely low rate affected by great kinetic difficulties of the reaction.

Thermodynamic Peculiarities of Metal Chemical Reduction

The electromotive force of the oxidation-reduction reaction is determined by the difference of oxidizer (Table 7.4) and reducer (Table 7.5) potentials. Values of oxidation-reduction potentials in real systems as a rule differ from standard values given in tables, according to component activities in the solution and pH of the medium.

To form a fine, highly dispersed product of reduction, it is necessary to provide a maximum difference of oxidation-reduction potentials of the reduced metal and the reducer. The ligands forming complexes with reduced metal ions are introduced into the solution to prevent metal hydroxides forming in the alkaline medium.

Metal reduction is possible thermodynamically if the oxidation-reduction potentials of the metal and the reducer are within the zone of their stability (shaded area I, II in Fig. 7.4, with the example of chemical precipitation of nickel). Soluble complexes of nickel are formed in the presence of ligands (citrate-ions) at pH = 7, and so the zone of nickel chemical precipitation is significantly expanded (I). Therefore, the nickel precipitation by means of formaldehyde becomes possible (Fig. 7.3, zone II).

Thermodynamic data allow the determination of a principal possibility of oxidation-reduction reaction performance, but do not provide any information concerning the rate of the process.

TABLE 7.4 Standard Oxidation-Reduction Potentials of Some Metals in Aqueous Solutions at 298 K

Electrode Reaction	Potential *E* (V(nhe))
$Ni^{2+} + 2e = Ni$	−0.23
$[Ni(NH_3)_6]^{2+} + 2e = Ni + 6NH_3$	−0.476
$Co^{2+} + 2e = Co$	−0.277
$Cu^{2+} + 2e = Cu$	+0.337
$[CuOOH]^{+} + 2e = Cu + HCOO^{-}$	+0.285
$[CuCH_3COO] + 2e = Cu + CH_3COO$	+0.276
$[Cu(NH_3)_4]^{2+} + 2e = Cu + 4NH_3$	−0.05
$[Cu(NH_3)_2]^{+} + e = Cu + 2NH_3$	−0.11
$Ag^{+} + e = Ag$	+0.799
$[AgNH_3]^{+} + e = Ag + NH_3$	+0.603
$[AgCH_3COO]^{+} + e = Ag + CH_3COO^{-}$	+0.643
$[AgCHCO]^{+} + e = Ag + HCOO^{-}$	+0.502
$[Ag(NH_3)_2]^{+} + e = Ag + 2NH_3$	+0.373
$AgCN + e = Ag + CN^{-}$	−0.04
$[Ag(CN)_3]^{2-} + e = Ag + 3CN^{-}$	−0.51
$Au^{+} + e = Au$	+1.692
$[AgCl_2]^{-} + e = Au + 2Cl^{-}$	+1.154
$[AgCl_4]^{-} + 3e = Au + 4Cl^{-}$	+1.002

TABLE 7.5 Oxidation-Reduction Potentials of Reducers in Aqueous Solutions [7]

Electrode Reaction	Potential *E*, V(nhe)
$H_2PO_2^{-} + 3OH^{-} = HPO_3^{2-} + 2H_2O + 2e$	−0.31–0.09 pH
$HPO_3^{2-} + 3OH^{-} = PO_4^{3-} + 2H_2O + 2e$	+0.14–0.09 pH
$H_3PO_2^{-} + H_2O = H_3PO_3 + 2H^{+} + 2e$	−0.499–0.06 pH
$H_3PO_2^{-} + H_2O = H_2PO_3^{-} + 3H^{+} + 2e$	−0.446–0.09 pH
$H_2PO_2^{-} + H_2O = HPO_3^{2-} + 3H^{+} + 2e$	−0.323–0.09 pH
$H_2PO_2^{-} + 2H^{+} + e = 2H_2O + P$ (red form)	−0.248–0.12 pH
$HCHO + 3OH = HCOO + 2H_2O + 2e$	+0.19–0.09 pH
$HCHO + H_2O = HCOOH^{-} + 2H^{+} + 2e$	+0.056–0.06 pH
$2HCHO + 4OH = 2HCOO + 2H_2O + 2e$	+0.32–0.12 pH
$BH_4^{-} + 8OH = BH_2^{-} + 6H_2O + 8e$	−0.45–0.06 pH
$N_2H_4 + 4OH = N_2 + 4H_2O + 4e$	−0.31–0.06 pH
$N_2H_4 + 2OH = 2NH_2OH + 2e$	+1.57–0.06 pH
$N_2H_5^{+} = N_2 + 5H^{+} + 4e$	−0.23–0.075 pH
$NH_2OH + 7OH = NO_3^{-} + 5H_2O + 6e$	+0.683–0.07 pH
$2NH_2OH + 4OH = N_2O + 5H_2O + 4e$	−0.21–0.06 pH

TABLE 7.5 Oxidation-Reduction Potentials of Reducers in Aqueous Solutions—cont'd

Electrode Reaction	Potential *E*, V(nhe)
$S_2O_4^{2-} + 4OH = 2SO_3^{2-} + 2H_2O + 2e$	+0.56–0.12 pH
$SO_3^{2-} + 2OH = SO_4^{2-} + H_2O + 2e$	−0.09–0.06 pH
$S_2O_3^{2-} + 6OH^- = 2SO_3^{2-} + 3H_2O + 4e$	+0.69–0.09 pH
$HSnO_2^- + H_2O + 3OH^- = Sn(OH)_6^{2-} + 2e$	+0.33–0.09 pH
$Ti^{3+} = Ti^{4+} + e$	−0.04
$Sn^{2+} = Sn^{4+} + 2e$	+0.151
$Cr^{2+} = Cr^{3+} + e$	+0.41
$Fe^{2+} = Fe^{3+} + e$	+0.771
$V^{2+} = V^{3+} + e$	−0.25
$R_2NHBH_3 + 3H_2O = R_2H_2N^+ + H_3BO_3 + 5H^+ + 6e$	*E*[a] −0.05 pH

[a]*E is dependent on nature of* R.

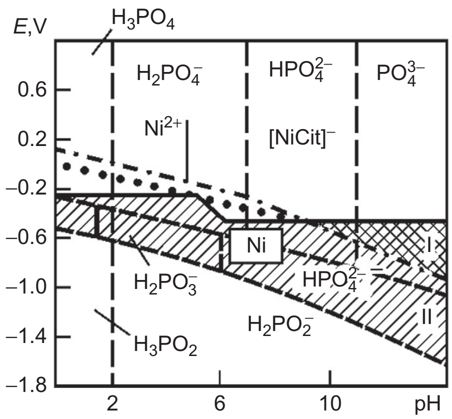

FIG. 7.4 Dependence of potential on pH: upper dashed line is balance of HPO_3^{2-}/PO_4^{3-}; point line is $2H^+/H_2$; solid line corresponds to Ni^{2+}/Ni (left part) and $[NiCit]^-/Ni$ (right part); upper dotted line corresponds to $H_2PO_2^-/HPO_3^{2-}$; lower dotted line corresponds to $H_3PO_2/H_2PO_3^-$ (left part) and $H_2PO_2^-/H_2PO_3^-$ (right part).

Kinetics of Solid-Phase Formation in Solution

The rate of metal chemical reduction from solutions is determined by kinetic factors, by a constant process rate, and the duration of the induction period. The slow reaction in the initial stage depends on the formation of stable particles of solid phase, which show an autocatalytic effect in the following stages. During the initial stage, the reaction proceeds by successive coarsening of nonstable fine particles of the solid product. For example, the crystallization of silver nuclei is performed step by step through the formation of successively coarsening clusters:

$$Ag^+ + Red \leftrightarrow Ag_1 + Red^+$$

$$Ag_1 + Ag^+ \leftrightarrow Ag_2^+$$

$$Ag_2^+ + Red \leftrightarrow Ag_2 + Red^+$$

$$Ag_m + Ag_n = Ag_{m+n}$$

where Red is the nucleus of reductant.

Clusters containing a small number of atoms are thermodynamically nonstable (the oxidation-reduction potential of atomic silver is 1.8 V (nhe) instead 0.799 V for the metallic silver phase). Metal particles become stable in the reaction

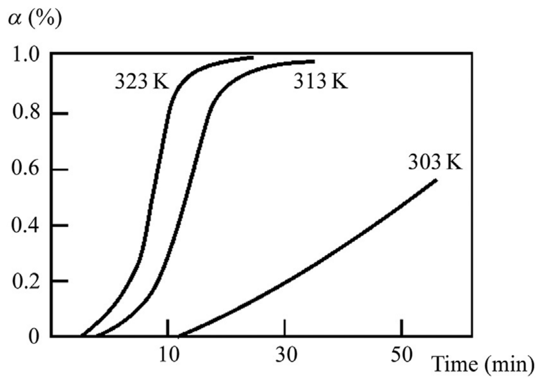

FIG. 7.5 Kinetic curves of Cu (II) reduction by formaldehyde.

medium after achieving 1–5 nm particle size. Assuming the number of growing metal particles to be unchanged after the induction period and the particles to have a spherical shape, the rate of their growth is constant and proportional to the surface S:

$$\frac{d\alpha}{dt} = k_1 S = k_2 S^{2/3}$$

where α is the reduced metal yield in the fractional part of unity; and k_1 and k_2 are the constants of reaction rate.

The integral form of the equation, which describes the initial sections of reaction kinetic curves (Fig. 7.5), is

$$\alpha = kt^3$$

where k is the integration constant and t is the duration of reaction.

In the case of diffusion control of metal reduction, the rate of particle growth is proportional to its radius and is described by the equation

$$\alpha = kt^{3/2}$$

where k is the rate constant (its value for temperature ranges from 313 to 323 K amounts to 1.66–1.67).

In the general case of stage formation of crystallization centers, the equation of kinetics of the metal crystallization process from liquid melt is developed:

$$\alpha = 1 - \exp\left(-kt^{\mathrm{n}}\right)$$

which after series expansion is reduced to $\alpha = kt^n$ and describes the reduction of metal ions (copper and silver) from aqueous solutions by various reducers, and 'n' can vary from 1.5 to 10. The initial stages of the reduction reaction are followed by the period when the process rate depends not only on the value of the particle surface, but on the concentration of ions of the metal ($C_{M^{n+}}$) and reducer (C_{Red})

$$\frac{d\alpha}{dt} = kC^{\mathrm{a}}_{\mathrm{M}^{\mathrm{n}+}} C^{\mathrm{b}}_{\mathrm{Red}} S$$

where α is the reduced metal yield as the fraction of unity, and 'a' and 'b' are constants.

In the case of reducer excess in the solution, the following proportion

$$kt_1 = 0.5\ \ln\left(\frac{1+\alpha^{1/3}+\alpha^{2/3}}{\left(1-\alpha^{1/3}\right)^2}\right) + \sqrt{3}\mathrm{arctg}\left(\frac{2\alpha^{1/3}+1}{\sqrt{3}}\right)$$

describes the kinetic reduction curves for some metals rather well; copper is reduced by formaldehyde; tellurium is reduced by hydrazine; silver is reduced by ascorbic acid and hydroquinone, etc.

By chemical reduction, metals can be produced as sols or powders depending on the chosen method. Particle shape and size are controlled by the nucleation rate. In the case of sol formation, the reduction is controlled by the formation of new metal nuclei; the rate of the development of these nuclei is small. In the case of powder production, the conditions for nucleation are less favorable, but the rate of particle growth is appreciable. Also, aggregation of fine particles into larger agglomerates takes place.

A momentary formation of finely divided reduction product is realized at large values of the difference of chemical reagent's oxidation-reduction potentials and in the absence of the autocatalytic effect of the newly formed surface of metallic particles. Metallic colloids are usually formed at the maximum allowable concentration of the reducer and rather low concentration of metal ions. Ligands and protective colloids are introduced into the solution for stabilization of metal sols. Sols of various metals in borohydride solution are produced at elevated temperature and in the presence of a protective colloid (for example, gelatin). In such a way, sols of Ag, Au, Pd, Os, Rh, Pt, Cu, Sn, Tl, Te, Co(B), and Ni(B) can be synthesized. In the case of noncomplex metal ions, sols are homogeneous and superfine. In the case of the reduction of complexes, the degree of sol homogeneity decreases with the growth of their strength.

The growth rate and size of powder particles are controlled by changing the nature and concentration of the reducer, the concentration of precipitated metal ions, the solution temperature, and the concentration of complexing agents or surfactant. Monodisperse powders are formed during cooling by thorough and intensive stirring of the reaction medium. To reduce particle size, the solution temperature is decreased and complexing agents and surfactants are introduced.

Application of Chemically Reduced Powders and Sols

Metal powders and sols are used as catalysts in many organic reactions. The catalyst's activity increases by deposition on a carrier. Monodisperse powders with uniform distribution of the catalyst on the surface are produced by applying chemically reduced metal sols on the carrier particle surface. There are several methods of sol deposition:

- A metal colloid is precipitated by a chemical method and a metallized dispersed carrier is produced in a brief contact of the metal colloid with the carrier particles as a result of colloid adsorption. The method is possible if the rate of colloid adsorption is higher than its aggregation rate.
- Colloid production directly in the presence of the carrier is preferable. Carrier particles impregnated by a salt of reducible metal are put into the reducer solution and colloid metal particles are formed on the carrier surface. $Al(OH)_3$, TiO_2, Si, MgO, $CaCO_3$, ZrO, and carbon fiber material are used as carriers.
- Selective precipitation of chemically reduced metal on the carrier is performed by preliminary activation of the carrier surface by microcrystals of silver or palladium. Subject to conditions of preliminary activation and precipitation, it is possible to produce particles with controlled dispersivity, differing by thermal stability in relation.

AUTOCLAVE PRECIPITATION OF METAL POWDERS

This method includes processing of a metal salt solution (solutions of metal hydroxide or carbonate pulp are rather rare) by a reagent-reducer at elevated temperatures and pressures [1,2]. The autoclave scheme for hydrogen reduction is shown in Fig. 7.6. Some companies use the autoclave method to produce nickel, cobalt, and copper powders. The processing of laterite deposits of nickel and cobalt ore high in iron is accomplished through pressure leaching with sulfuric acid in Moa, Cuba. The nickel and cobalt sulfide concentrates so produced are the principal feed to the Sherritt International Refinery in Fort Saskatchewan, Alberta, Canada. The Sherritt refining process includes leaching at elevated temperature and pressure in the leach autoclaves. The Sherritt process is adaptable to handle a wide variety of different feedstocks. It has been adopted by Western Mining Corporation in a refinery at Kwinana, Western Australia. The Sherritt nickel and cobalt refining processes are further discussed in Chapters 21 and 22.

Organic reagents (glucose, formaldehyde, hydrazine, etc.) are used as reducers, but they contaminate the final solution and powder by their dissociation products. Gaseous precipitators, that is, sulfur oxide, carbon monoxide, and hydrogen are more effective. Hydrogen is the most widely used gaseous reducer; it is an effective, readily available reagent that does not contaminate the solution.

According to the experience of autoclave plants, hydrogen is produced by natural gas conversion. Possible impurities (CO_2, CO, H_2O and H_2S) lead to decreasing hydrogen activity, increasing total pressure in the autoclave, contaminating the metal powder, and the formation of carbonate ions in the ammonia media. This fact increases ammonia consumption and changes the molar ratio NH_3 (free) and $(NH_4)_2CO_3$, which is a kinetically important parameter.

Sulfur dioxide is a well-known reagent, but it is toxic, has limited critical parameters, and leads to the accumulation of sulfuric acid in the solution and powder contamination. This reagent, like carbon monoxide, is rather useful for precipitation of metal powders from alkaline and ammonium media. However, carbon monoxide possesses some shortcomings: it is toxic, explosive, difficult to synthesize, and expensive as well as changes the acid-basшc equilibrium.

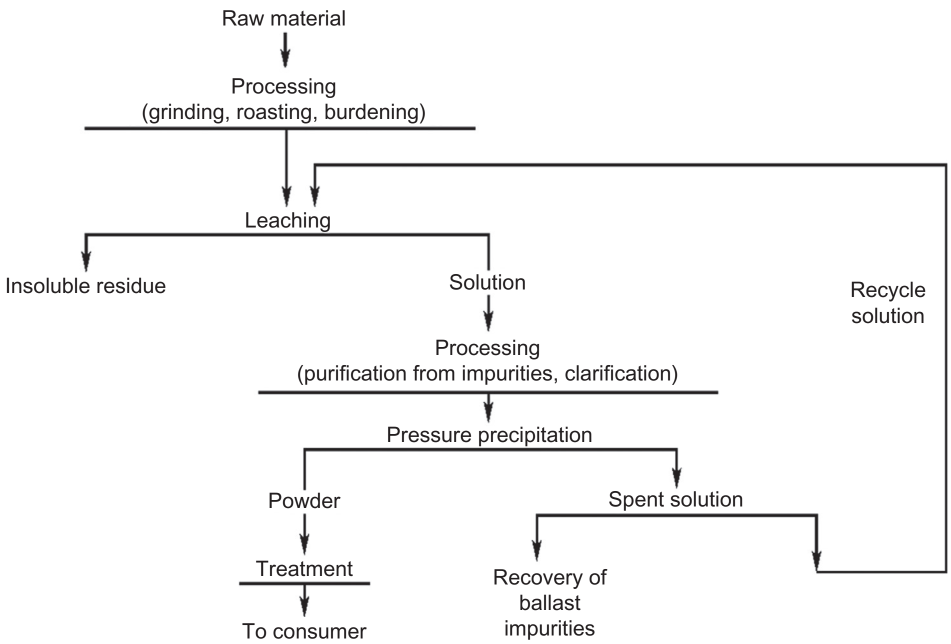

FIG. 7.6 Flowsheet for metal powder production by the autoclave method.

Theory of Autoclave Metal Ions Reduction

The reaction of metal ion reduction by hydrogen

$$M^{n+} + \frac{n}{2}H_2 \rightarrow M + nH^+$$

will proceed to the right until the hydrogen potential (E_{H_2}) is more electronegative than the potential of the metal ion (E_M^{n+1}). The limiting degree of metal ion reduction is achieved at the moment when $E_{H_2} = E_M^{n+1}$.

The moving strength of the summary reaction is defined by the difference of potential boundary phases M^{n+}/M and H^+/H_2. These potentials are determined per the Nernst equation [15]:

$$E_{M^{n+}/M} = E^0_{M^{n+}/M} + \frac{RT}{nF}\ln(a_{M^{n+}}) \tag{7.1}$$

where $E^0_{M^{n+}/M}$ is the standard potential of this boundary phase (standard conditions: $T = 298$ K, activity of metal ions in solution g-ion/L); n is the metal ion charge; R is the universal gas constant; T is the temperature; F is the Faraday constant; is the activity factor for M^{n+}; and $C_{M^{n+}}$ is the concentration of metal ions.

For hydrogen, as its standard potential is zero by definition, the boundary potential is connected with hydrogen-ion activity (a_{H^+}) in the following manner:

$$E_{H^+/H} = 0 + \frac{RT}{2F}\ln\left(\frac{(a_{H^+})^2}{p_{H_2}}\right) = \frac{RT}{F}\ln(a_{H^+}) - \frac{RT}{2F}\ln(p_{H_2})$$

where p_{H_2} is the hydrogen gas pressure.

The substitution of a negative logarithm of hydrogen-ion activity by the conventional symbol pH gives:

$$E_{H^+/H} = -\frac{2.303RT}{F}\left(\mathrm{pH} + \frac{1}{2}\lg(p_{H_2})\right) \tag{7.2}$$

The equilibrium activity $a_{M^{n+}}$ and concentration of metal ions in the solution are calculated by means of joint decision of Eqs. (7.1) and (7.2). The intermediate transformations lead to the equation:

$$\lg(a_{M^{n+}}) = -n \cdot \mathrm{pH} - \frac{n}{2}\lg(p_{H_2}) - \frac{nF \cdot E^T_{M^{n+}}}{2.303 \cdot RT}$$

where is the standard potential for metal with correction with respect to temperature T.

This equation is acceptable for conditions in dilute solutions (with ion concentration below 0.1 mol/L), in which the activity factor for M^{n+} is near 1.0.

In concentrated solutions, it is necessary to take into account the bond of the activity factor for M^n with the ionic force of solution and, after that, calculate the equilibrium concentration of metal ions $C_{M^{n+}}$.

$$\lg(C_{M^{n+}}) = -n \cdot \mathrm{pH} - \frac{n}{2}\lg(p_{H_2}) - \frac{nF \cdot E^{T}_{M^{n+}}}{2.303 \cdot RT} - \lg(f_{M^{n+}})$$

The values of activity factors are given in the reference literature; for sulfate media the following equation is valid:

$$f_{M^{n+}} = 0.056(\lg \mu)^2 + 0.016$$

where μ is the ionic force of solution equal to $\frac{1}{2}\sum n_i^2 \cdot c_i$; c_i is the concentration of ith ions, g mol/L.

To denote X, the share of reduced metal ions (M^{n+1}), the degree of its reduction is calculated according to the equation:

$$E^0_{M^{n+}/M} - \frac{0.0592}{n}\lg(1 - X) = 0.05921 \cdot \lg n \cdot X.$$

As the influence of pH is most significant, the alkali solutions are preferable for the precipitation of electronegative metals. In the presence of ammonia and other complex ligands, metal ions and complex ions with a different number of addenda are in the solution. For this case, the potential of the metal ion (E_{M_K}) is calculated as follows:

$$E_{M_K} = E_{M^{n+}/M} - \frac{RT}{nF}\ \ln\sum_0^m \beta_i \cdot a^i_{i_{free}}$$

where $E_{M^{n+}/M}$ is the potential of the metal ion in the solution with the same concentration without complex ligands; i is the number of addenda in the complex; β_i is the stability constant of the complex that contains i complexing groups; $a^i_{i_{free}}$ is the efficiency of excess complexing ligands; and m is the coordination number.

The pH value of the ammonia solution containing free ammonia and ammonium salt is calculated as:

$$\mathrm{pH} = pK^2\beta + \lg\ a_{NH_3\,free} + \lg\frac{a_{NH_4^+} \cdot a_{OH^-}}{a_{NH_4OH}} - \lg\ a_{NH_4^+}.$$

The concentration of free ammonia in the solution is determined by pH-potentiometric titration.

Thermodynamically, the reduction of many metal ions by hydrogen is possible already at 298 K and $p_{H_2} = 0.1$ MPa. However, the implementation is first determined by kinetic factors. The rate of autoclave precipitation of metals is defined by hydrodynamics, pressure of the gas reducer, temperature, and solution (pulp) composition.

Hydrodynamics

Effective saturation of the liquid phase by the reaction gas and uniform density of the formed pulp with minimal energy consumption are achieved by sufficiently intensive stirring. Mechanical mixers that provide mass exchange by means of configuration-aerated units or rotation intensity are used; the latter is limited by the increasing energy consumption of mixer drives, the complexity of tightening units for their inlet into the reactor, and intensive foaming. Aeration units are more effective, but they lose the effect of the presence of powder surface deposits.

Parameters and Components of the Process

Reaction Gas Pressure

When the gas pressure is increased, the concentration of dissolved gas diluted in the pulp proportionally increases, and that accelerates the process. Critical pressures are limited by design reasons, the extra cost for gas compression, and the complexity and danger of plant servicing. The partial pressures of reducing gas vary between 0.05 and 3.0 MPa.

Temperature

The use of a higher temperature is particularly justified by effective aeration, but it involves a greater consumption of energy, more risk of corrosion, and greater pressure in the autoclave, especially if volatile reagents such as ammonia are used. Also, the composition of the solution may change because of hydrolysis, the sintering of powder particles, and the decreasing solubility of the salts. Furthermore, the operation of equipment is more difficult. Usually the precipitation is performed in the temperature range from 400 to 500 K.

Solution Composition

Ammonia, sulfuric acid, hydrochloric acid, and organic media are used; attention is increased to the processing of metal hydroxide pulps. Process rate and powder yield per unit volume increase with the growth of precipitable metal content on the condition that a reducing gas deficit is avoided. The upper limit is determined by solution stability (restricted solubility and metal salt hydrolysis); the rheological properties of pulps and the effectiveness of total mass exchange become limiting factors in pulp processing. The lower limit is determined by economic indexes of the process.

Free acid in the solution complicates the corrosion condition and decelerates the rate of reducing gas molecule activation. A minimal acidity sufficient for hydrolysis prevention at elevated temperatures and introducing buffering salts (ammonia salts and alkali metal salts are preferable) are recommended. When using ammonia media, the concentration of free ammonia is chosen, taking into account the provision of the stability of diamine complexes of the $(M(NH_3)_m)^{n+}$ type, where $m = 2.2$–2.5. In the case of ammonia excess, the total pressure in the autoclave and the loss of gas vapor mixture are increased.

Depending on the initial raw composition, the solutions used for autoclave precipitation contain various impurities, especially when using acid closed-circuit technologies with respect to the solution. With organic ammonia-based solvents, which provide highly selective leaching, the problem of coprecipitation of metallic impurities arises (for example, in nickel powder production-cobalt and copper coprecipitation). Therefore, a prior solution purification is required.

Accumulated impurities cause increased reducing gas consumption, change the solution properties (kinematic viscosity, solubility, and pH), and lead to powder contamination. To exclude the ingress of suspended solids, the initial solution must be thoroughly filtered.

During metal precipitation, the process of ion reduction is developed by a homogeneous mechanism of reacting gas activation. The ions with alternating valence play a significant part. They can accelerate the process or oxidize the intermediate ions formed with lower valence of the precipitated metal while causing an increase of reaction gas consumption and a lengthening of the precipitation process.

Some impurity anions form insoluble compounds with metal ions, which are reduced to intermediate valences. For example, the formed copper halides are insoluble, contaminating copper powder and decreasing its yield.

Seed

The effect of introduced dispersed solid substances (similar powder, graphite, carbides, borides, etc.) appears during the metal precipitation process, the mechanism of which obeys the regularities of heterogeneous catalysis per reducing gas molecule activation (for example, under hydrogen precipitation of nickel and cobalt). Introduction of a seed not only accelerates the process but also helps to control particle size and cover seed particles with a precipitated metal layer to obtain composite powders. The autoclave cladding technique provides the treatment of materials with different particle size, layer thickness control, and a high rate and absence of toxic emissions.

Surfactants

Insignificant additions of organic substances (polyacrylic salts, anthraquinone, and its derivatives) eliminate powder deposits on the autoclave's inside surface; influence the powder dispersity, shape, and texture of powder particles; provide uniform covering of the precipitated metal on the surface of inert seeds; and accelerate the process. But then, the surfactants introduced and the products of their destruction contaminate the powder with carbon.

Polyacrylamide (PAA) surfactant (0.4 g/L) introduced into the initial solution of the autoclave cupper powder production prevents the skull formation on the inner surface wall of the reactor and increases the metal yield. An 80–85 wt% minimum copper yield was obtained in the following conditions: initial copper concentration of 1.0–1.2 wt%, $(NH_4)_2SO_4$ of 1.2–1.5 wt%, sulfuric acid up to 0.15–0.25 wt%, at 408–418 K temperature, and under 2.5–2.8 MPa pressure. While in the autoclave process without PAA, a significant part of the powder (up to 20%–30% of the copper mass yield) precipitates on the inner surface wall of the reactor. However, the effect is reached by the surfactants content COO^- carboxyl group and along with this, they must be polymeric and stable up to 673–700 K.

The mechanism of the antiskull effect in the presence of the carboxyl containing surfactant may be described in the following way:

1. During dissolution of polymer compounds due to micelle formation, a high dispersity two-phase system is formed.
2. In weak acidity copper solutions, carboxyl compounds of the surfactant form a complex with copper ions. Due to the development of chemisorption processes, a micelle structure of dissolved surfactant promotes the appearance of numerous centers with high copper content that accelerate the formation of cupric ions by interaction with hydrogen.
3. The appearance of a metallic phase by autoclave copper precipitation proceeds through a phase of cupric ion formation with their further disproportionation; the formation rate and cupric concentration are proportional to the initial copper concentration.

Therefore, the conditions for nucleation and particle formation of copper directly in the solution volume are created; consequently, the depositions on the reactor inner surface wall are reduced.

Autoclave Processing of Hydroxide and Metal Carbonate Pulps

In general, the method of autoclave precipitation of metal powder includes the following stages:

- Gas dissolution and saturation of the aqueous phase by it.
- Adsorption of the gas on the seed surface and its activation with the formation of the intermediate metal hydride (heterolytic mechanism):

$$MX_{n-1}^{m-n} + H_2 \leftrightarrow MX_{n-1}H^{m-n} + H^+ + X^-$$

- Gas activation by metal ions with alternating valence (homolytical mechanism):

$$2MX_n^{m-n} + H_2 \leftrightarrow 2MH^{m+} + 2nX^-$$

- Electrochemical reaction of metal ion reduction.
- Reaction products transfer to solution volume.

Hydroxides or carbonates of metals are semiproducts of their extraction from low-grade raw materials or are formed during neutralization of technological overflow solutions. Direct autoclave processing of pulps provides a greater volume of powder and is useful for the production of powders with special properties (catalytically active powders, powders with narrow size distribution, powders with roundish shape particles, and ultrafine powders). Solid-phase composition and morphology, temperature, hydrogen pressure, catalyst consumption, and metal content in the aqueous phase are the main factors influencing the choice of pulp processing parameters.

So far as by hydrothermal conditions, the basic metal carbonate undergoes destruction and is transformed into metal hydroxide; the theoretical positions are considered using hydroxide as an example.

The summary reaction $M(OH)_2 + H_2 \rightarrow M + H_2O$ consists of two subreactions:

$$M^{2+} + 2e \rightarrow M^0, E_1^0 \text{ and } 2H^+ + 2e \rightarrow H_2, E_2^0$$

The difference in potential for them is the following:

$$\Delta E = E_1^0 - E_2^0 - 2.303\frac{RT}{2F}\left(2\lg K_W - \lg Lp_{M(OH)_2}\right)$$

where $K_W = [H^+][OH^-]$; and $Lp_{M(OH)_2}$ is the solubility product of $M(OH)_2$ and $Lp_{M(OH)_2} = [M^{2+}][OH^-]^2$.

Temperature, dissolved hydrogen concentration, and metal hydroxide solubility are the characteristic parameters of this system.

The process is carried out at a temperature range of 470–520 K, $p_{H_2} = 2 - 3$ MPa over 1–1.5 h using pulps that contain 25%–40% of solid phase. The powder properties depend on the pulp composition, shape, and size of solid phase particles and process parameters. The contents of carbon dioxide and ammonia influence the value of apparent density; the powder particle size depends on metal concentration, temperature, and hydrogen pressure.

If semiproducts are used, then, for pure powder production, it is useful to dissolve them to purify the solution and to produce secondary carbonate after solution redistillation.

Autoclave Precipitation of Metal Powders from Organic Phases

This method of powder production allows producing ultrafine powders with a developed surface and widening the choice of raw material base by implication of low-grade raw material, which can be processed by a leaching-extraction process. This method has been developed for the production of nickel, cobalt, copper, and noble metal powders.

The organic phase should be thermally stable with low solubility in water and be quantitatively regenerated during autoclave processing. Carboxylic acids and derivatives on their base and chelate compounds relate to similar reagents. Tertiary carboxylic acid $(CH_3)_3CCOOH$ is most thermally stable up to temperatures of 520–570 K. It is used in a mixture with diluent in the ratio of 1 to 2.

The advantages of autoclave metal precipitation from organometallic phases are their noncorrosiveness to stainless steels; significant decreasing of total pressure in the unit (lower than vapor pressure, below 0.5 MPa at 570 K); prevention of particle flocculation; property features of powder (roundish shape of particles, and fine particle sizes within several micrometers); organic phase regeneration; and a more compact flow diagram of powder production in an extraction-precipitation cycle.

But commercial application is restricted by the limited choice of available reagents that are stable at 348–483 K; their unavoidable partial destruction and contamination of the powder by carbon, sulfur, and phosphorus; fire risk; difficulties in separation of produced powder pulp; and the high power consumption of the process (owing to lean per metal solutions and elevated temperatures).

Equipment

For hydrogen precipitation of metals, the vertical and horizontal autoclaves with mechanical stirring are used; the latter type is predominantly used for scale production. The autoclave body is made of carbon steel clad inside with a corrosion-proof material: stainless steels or high alloyed steels by processing of ammonia, organic media, and pulp of metal hydroxides and titanium by processing of sulfurous and chloride media).

The autoclaves are equipped with a hatch for inside inspection and maintenance, fittings for installation of gauges for process characteristics control (thermocouples and pressure gauges), safety fittings (explosive valve, check valve), and tubing for the supply of solution, gas feeding, and discharge of the products produced. Gases and the initial solution are supplied from the top into the autoclave; powder pulp is discharged through a dipping tube. The diagrammatic pattern for a horizontal autoclave is shown in Fig. 7.7.

Autoclaves for powder precipitation work parallel in periodic mode. Continuous mode is also possible, but it is complicated because of the possibility of autoclave "obliteration," the risk of hydrogen leakages, and the complexity of unit operation for solution supply and especially for powder pulp discharge. Also, there is difficulty with control of the particle size and powder particle density. The load to stirrers drives is controlled, especially at the end of the process when pulp density increases and is quickly laminated. Overheating and shutdown of the stirrer is an emergency situation. Stirrer drives are equipped by guard relay actuators for thermal protection. Autoclaves are shut down regularly for inspection and maintenance with particular attention paid to the condition of the protective cladding and the welds.

Self-evaporators are capacitive devices with a conical bottom, which are equipped with a fitting for transfer of the solution and release of the gas-vapor phase. They are made of a corrosion-proof material (like the autoclave) and operate at atmospheric pressure. One self-evaporator is connected with two or three autoclaves as the duration of powder pulp processing and discharge is much smaller than the duration of the powder producing cycle in the autoclave. A double-pipe heat exchanger is the most popular type; it is manufactured from high-alloyed steel or titanium. Heat transfer efficiency

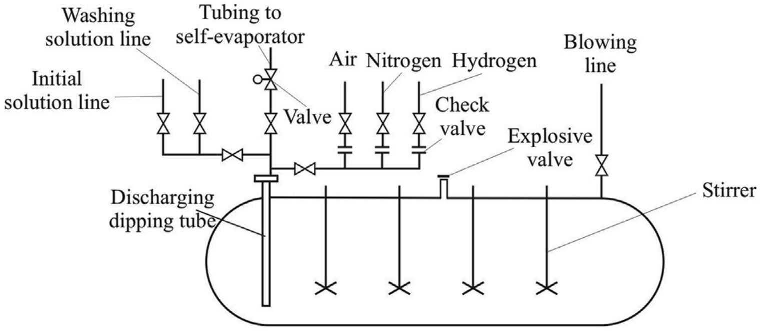

FIG. 7.7 Diagrammatic scheme for hydrogen reduction in an autoclave.

deteriorates in the long run due to the precipitation of basic salts of metals and calcium sulfate. The rate of deposit formation depends on solution composition and temperature. Alkaline salts and sulfate deposits are dissolved by the spent solution and by nitric acid, respectively.

Centrifugal pumps are used for transmission of the solutions; their supply into vessels operating under pressure is carried out by pumps of a plunger type. The amount of the solution supplied is controlled by the duration of pump operation and by prior measurement of the solution supply rate.

Discharging units of the autoclaves, the self-evaporator, and the line for powder pulp transportation are subject to strong erosion, so they should be changed periodically. Discharging lines are designed with the minimum number of elbows. Relief valves are changed from time to time.

The equipment used for powder treatment (drying, screening, mixing and stirring, milling, and packing) typical for powder metallurgy is described in Chapter 14.

Safety Engineering

Autoclaves for reduction are located in a separate building with easy dropping cover and big window openings [10]. The hydrogen content in the atmosphere in the building is controlled continuously [10–13].

The process gases are supplied into the autoclave via the separate gas tubing; the line for each gas has two faucets and a check valve allowing unused lines to switch off. The hydrogen pressure lines are controlled according to rules [16]. The whole system, including autoclaves, is purged with nitrogen ($p=0.11$ MPa) before hydrogen or air is let in. The gas-vapor mixture is continuously blown away from self-evaporators to eliminate hydrogen accumulation to critical concentration. Oxygen content in the gaseous phase after venting of equipment shall not exceed 2%. Hydrogen-containing gases from the autoclave and the self-evaporator are passed through a drip pan and released via a chimney out of the building area. Welding works have to be performed only after equipment shutdown, its purging, and ventilation of the building with continuous monitoring of the hydrogen content in the air.

Autoclave operation must be largely automated and be carried out by remote control. Control valves, electric drives of autoclaves stirrers, ventilation systems, and pumps must have explosion-proof performance. Intermittent inspection, monitoring, high maintenance culture, and personnel experience are required.

REFERENCES

[1] Kiparisov SS, Padalko OV. Equipment in powder metallurgical enterprises. Metallurgy: Moscow; 1988.
[2] Kakovsky IA, Naboychenko SS. Thermodynamics and kinetics of hydrometallurgical processes. Alma-Ata: Nauka; 1986.
[3] Chopra KL. Thin film solar cells. New York, London: Plenum Press; 1983.
[4] Naboychenko SS, editor. Handbook of NonFerrous Metal Powders. Moscow: Metallurgia Publishers; 1997 (in Russian).
[5] Sircar S. Pressure swing adsorption technology for hydrogen purification. In: Zhou L, editor. Adsorption. World Scientific; 2006. p. 29–45.
[6] Li B, He G, Jiang X, Dai Y, Ruan X. Pressure swing adsorption/membrane hybrid processes for hydrogen purification with a high recovery. Frontiers of Chemical Science and Engineering 2016;10(2):255–64.
[7] Elseviers W, Hassett PF, Navarre JL, Whysall M. 50 Years of PSA technology for H_2 purification, Des Plaines, IL: UOP LLC, A Honeywell Company UOP LLC; 2016.1–5. [other] https://www.uop.com/?document=psa-50-paper&download=1.
[8] US New Jersey Department of Health and Senior Services. Hazardous Substance Sheet. Hydrogen. Revision: July 2016. < http://nj.gov/health/eoh/rtkweb/documents/fs/1343.pdf> [Accessed 31 January 2018].
[9] Leiby SM. Options for refinery hydrogen. Menlo Park, CA: Process Economics Program, SRI International; February 1994 Report No. 212.
[10] Determination of explosion-proof and fire-safe categories of rooms and buildings. 2017. < http://www.nj.gov/health/eoh/rtkweb/documents/fs/1010.pdf > [Accessed 30 December 2017]
[11] Baratov AN, Korolychenko AJ, editors. Fire- and explosion-proof of matters and materials and means for their quenching. Moscow: Chimia; 1990.
[12] Guide to EU legislation and "Health and Safety" in the European PM industry. Shrewsbury: EPMA Publisher; 2009.
[13] Regulation for installation and safe operation for combustible, toxic and condensed gases. BOC Gases. European Development Centre, UK. <//www.boconline.co.uk/en/about_the_linde_group/about-boc/index.html>.
[14] Sviridov VV, Vorob'eva TN, Gaevskaja TV, Stepanova LI. Chemical deposition of metals from aqueous solutions. Minsk: Minsk University Publishing; 1987.
[15] Uhlig HH, Winston RR, Viley A. Corrosion and corrosion control. New York: John Wiley & Sons; 1985.
[16] PRXAIR. Hydrogen, compressed safety data sheet P-4604. Revision data: Oct 15, 2016. <http://www.boconline.co.uk/internet.lg.lg.gbr/en/images/tg-8360-hydrogen-v1.3410_84488.pdf>